SE 226 – LAB#2 2025-2026 SPRING

Aim: Flow control: selection statements , looping statements.

General Instructions

Create a Python project containing **lab3.py** and a C++ source file **lab3.cpp**. Complete all tasks in **both** programming languages. The problem names are provided as references — feel free to look them up online if you need further clarification.

Task 1 (Hailstone Sequence (Collatz Conjecture)

Take a **positive integer** from the user (greater than 1). Apply the following rule repeatedly until the value reaches **1**:

• If the current number is **even**, divide it by 2.

• If the current number is **odd**, multiply it by 3 and add 1.

Print each number in the sequence on the same line separated by **' → '**. At the end, print the total number of steps taken to reach 1.

Hint: Use a while loop and an if/else statement. No arrays or lists needed — just track the current value and a counter.

SAMPLE OUTPUT

```
Please enter a positive integer greater than 1: 6
6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1
Total steps: 8
```

SAMPLE OUTPUT (n=27)

```
Please enter a positive integer greater than 1: 27
27 → 82 → 41 → 124 → 62 → 31 → 94 → 47 → 142 → 71 →
214 → 107 → 322 → 161 → 484 → 242 → 121 → 364 → 182 →
91 → 274 → 137 → 412 → 206 → 103 → 310 → 155 → 466 →
233 → 700 → 350 → 175 → 526 → 263 → 790 → 395 → 1186 →
593 → 1780 → 890 → 445 → 1336 → 668 → 334 → 167 → 502 →
251 → 754 → 377 → 1132 → 566 → 283 → 850 → 425 → 1276 →
638 → 319 → 958 → 479 → 1438 → 719 → 2158 → 1079 → 3238 →
1619 → 4858 → 2429 → 7288 → 3644 → 1822 → 911 → 2734 →
1367 → 4102 → 2051 → 6154 → 3077 → 9232 → 4616 → 2308 →
1154 → 577 → 1732 → 866 → 433 → 1300 → 650 → 325 → 976 →
488 → 244 → 122 → 61 → 184 → 92 → 46 → 23 → 70 → 35 →
106 → 53 → 160 → 80 → 40 → 20 → 10 → 5 → 16 → 8 → 4 → 2 → 1
Total steps: 111
```

Task 2 (FizzBuzz Counter)

Take a positive integer **n** from the user (between **10 and 100** inclusive). If the user enters a value outside this range, keep asking until a valid value is given. Then iterate through all integers from **1 to n** and apply the following rules:

- If the number is divisible by both 3 and 5 → print **FizzBuzz**
- Else if divisible by 3 → print **Fizz**
- Else if divisible by 5 → print **Buzz**
- If the number is divisible by 7 → **skip it entirely** (use **continue**)
- Otherwise → print the number itself

After the loop, print a summary: how many **Fizz**, **Buzz**, and **FizzBuzz** values were printed.

Hint: Use a for loop with range(), elif chain, and the continue statement for the divisible-by-7 skip. Use a separate while loop for input validation.

SAMPLE OUTPUT (n=20)

```
Please enter a number between 10 and 100: 5
Invalid input. Please enter a number between 10 and 100: 110
Invalid input. Please enter a number between 10 and 100: 20
1
2
Fizz
4
Buzz
Fizz
(7 is skipped)
8
Fizz
Buzz
11
Fizz
13
(14 is skipped)
FizzBuzz
16
17
Fizz
19
Buzz
--- Summary ---
Fizz count : 5
Buzz count : 2
FizzBuzz count: 1
```

Task 3

Take a number from the user between 3 and 9 (inclusive). Print the following pattern using at most two loop structures.

Note: The top half has rows 1 to n, and the bottom half has rows n-1 down to 1. Each row contains k consecutive '' characters, where k is the current row's level.*

SAMPLE OUTPUT (n=4)

```
Please enter a number between 3 and 9: 4
*
**
***
****
***
**
*
```

SAMPLE OUTPUT (n=5)

```
Please enter a number between 3 and 9: 7
*
**
***
****
*****
******
*******
******
*****
****
***
**
*
```

Hint: For each row i (1 ≤ i ≤ 2n-1), determine the current 'level' k = n - |n - i|. Print k consecutive '' characters.*

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*Please complete all tasks in both **Python and C++**. Upload your `lab3.py` and `lab3.cpp` files into your **GitHub account**, then share the link to your repository/folder via **Blackboard**.*